SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 4.1: Greenhouse Effect and Climate Change — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 4.1: Greenhouse Effect and Climate Change
Driving Question	DRIVING QUESTION: Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable — and what can the physics of our atmosphere tell us about how to respond?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Is the Greenhouse Effect? — Anchoring Phenomenon Introduction**	Students examined the anchoring phenomenon: photographs and data showing the dramatic retreat of Mount Kenya's Lewis Glacier since 1900, paired with rainfall anomaly charts from ASAL (Arid and Semi-Arid Land) farming counties such as Kitui and Marsabit. They observed that global average temperatures have risen alongside rising CO_2 concentrations, and noted the visible shrinkage of glacial ice over decades. Teacher note: Spend adequate time here — students' initial confusion and curiosity about the phenomenon is the engine for all subsequent lessons. Do NOT resolve the phenomenon yet.	Students learned that the greenhouse effect is the natural process by which certain atmospheric gases — carbon dioxide (CO_2), water vapour (H_2O), methane (CH_4), and nitrous oxide (N_2O) — trap outgoing infrared (heat) radiation, keeping Earth's surface warmer than it would otherwise be. Without any greenhouse effect, Earth's average surface temperature would be approximately −18 °C instead of the habitable +15 °C. Students also learned the names and relative atmospheric concentrations of the four major greenhouse gases and began to distinguish the natural greenhouse effect from human-amplified climate change. Teacher note: Students frequently conflate the ozone hole with the greenhouse effect — flag this as a common misconception to revisit formally in Lesson 6.	The anchoring phenomenon (melting glaciers + unpredictable rains) is explained at the introductory level by the fact that a thickened 'blanket' of greenhouse gases is trapping more heat in Kenya's atmosphere than in pre-industrial times. The glacier retreat and rainfall disruption are both downstream consequences of this enhanced heat retention. At this stage students can offer only a partial causal explanation; the DQB should capture their remaining questions so the explanation deepens across all seven lessons. Teacher note: Record every student question on the Driving Question Board verbatim — these become the lesson-by-lesson success criteria for the unit.
Lesson 2 **Earth's Radiation Budget — Balancing Incoming and Outgoing Energy**	Students analysed simplified radiation budget diagrams and real NASA/NOAA data tables showing how 100 units of incoming solar energy are distributed: approximately 30 units reflected by clouds,	Students learned the concept of Earth's radiation budget: for a stable climate, outgoing longwave radiation emitted by Earth must balance incoming shortwave solar radiation. Key quantities introduced	The partially explained phenomenon: Mount Kenya's glaciers are melting because the global radiation budget is currently out of balance — more solar energy is being retained by the atmosphere than is being

	aerosols, and Earth's surface (albedo); roughly 19 units absorbed by the atmosphere; and about 51 units absorbed by Earth's surface. They compared the albedo of the Lewis Glacier's white ice surface with the darker rocky surface now exposed as the glacier retreats, noting the ice–albedo feedback visually. Teacher note: Use a physical demonstration — shine a torch on white paper vs. black paper and have students feel the warmth with the back of their hand — to make albedo visceral before introducing quantitative data.	include solar irradiance (~1361 W/m²), the planetary albedo (~0.30), and the Stefan–Boltzmann law at a conceptual level (hotter objects emit more radiation). Students also learned that any imbalance — more energy in than out — results in planetary warming, which is the current observed state. Teacher note: The Stefan–Boltzmann equation ($P = \sigma A T^4$) is introduced here conceptually; algebraic manipulation is formalised in Lesson 3. Ensure students can articulate the idea of energy balance in words before moving to equations.	radiated back to space. As the glacier retreats, the exposed dark rock absorbs more solar radiation (lower albedo), accelerating local warming further. This ice–albedo feedback is a physics-grounded explanation for why glacier retreat can become self-reinforcing. Teacher note: Connect explicitly to the farmers' rainfall disruption by noting that a warmer atmosphere holds more water vapour, altering convection patterns over the Rift Valley and changing when and where rain falls.
Lesson 3 **How Does the Greenhouse Effect Actually Work? — Tracing Energy Through Kenya's Atmosphere**	Students traced the path of a photon of solar energy through a layered model of the atmosphere (troposphere, stratosphere, mesosphere). They observed a simulation or physical analogy — such as a sealed glass jar with a thermometer warmed under a lamp versus an open container — showing how an enclosed medium retains more heat. Data on atmospheric CO_2 concentrations from Mauna Loa (and equivalent African monitoring stations) were presented alongside temperature records, allowing students to observe the strong positive correlation. Teacher note: The jar demonstration is highly effective but requires careful scaffolding — ask students to predict temperatures before observing, then compare predictions to results to drive cognitive conflict.	Students learned the precise physics mechanism of the greenhouse effect: (1) Incoming shortwave solar radiation (visible and UV light) passes through greenhouse gases largely unimpeded and is absorbed by Earth's surface, warming it. (2) Earth's warmed surface re-emits energy as longwave infrared (IR) radiation. (3) Greenhouse gas molecules absorb this outgoing IR radiation because their molecular bond vibration frequencies match IR wavelengths — CO_2, CH_4, H_2O, and N_2O are IR-active molecules. (4) These excited molecules re-emit IR in all directions, including back toward Earth's surface, raising surface temperatures. Students applied the Stefan–Boltzmann relationship quantitatively in simple calculations. Teacher note: The molecular vibration explanation is the conceptual core of the unit — allocate extra time here and use a molecular model or animation to show CO_2 bending and stretching modes.	Students can now provide a mechanistic explanation of why Mount Kenya is warming: increased concentrations of IR-absorbing greenhouse gases mean more outgoing infrared radiation is intercepted and re-emitted downward, raising the energy input to the glacier surface above the threshold needed to melt ice. The physics directly links atmospheric composition to surface temperature change at the glacier. Teacher note: Push students to articulate the causal chain in their own words: more CO_2 → more IR absorption → more downward IR re-emission → more energy at glacier surface → melting. This chain should be visible on the DQB and in students' model diagrams.
Lesson 4 **Greenhouse Gases in Detail — Properties, Sources, and Global Warming Potential**	Students examined a comparative data table of the four main greenhouse gases — CO_2, CH_4, N_2O, and H_2O vapour — showing their current atmospheric concentrations (in ppm or ppb), pre-industrial concentrations, residence times in the atmosphere, and Global Warming Potential (GWP) values relative to CO_2 over a 100-year horizon. Kenyan-context sources were highlighted: charcoal burning	Students learned the physical and chemical properties that make a gas a potent greenhouse gas: it must be IR-active (possess a dipole moment change during molecular vibration) and persist long enough in the atmosphere to exert sustained warming. They learned specific GWP values: $CH_4 \approx 28–36\times CO_2$; $N_2O \approx 265–298\times CO_2$ over 100 years. Students also learned that water vapour is the most	Students can now explain why not all greenhouse gas emissions are equally urgent: a Kenyan farmer switching from charcoal to biogas reduces both CO_2 and CH_4 emissions, with the methane reduction having a disproportionately large near-term climate benefit due to its high GWP. This lesson adds nuance to the anchoring phenomenon explanation — the unpredictability of Kenyan rains is driven

	(CO_2), livestock (CH_4 from enteric fermentation in Maasai cattle), rice paddies and sewage in Kisumu (CH_4), and agricultural soils (N_2O). Students used GWP values to calculate the CO_2-equivalent warming impact of a Kenyan smallholder farm's methane emissions. Teacher note: The GWP calculation is an excellent numeracy integration point — ensure students understand GWP as a comparative index, not an absolute temperature value.	abundant greenhouse gas but acts primarily as a feedback amplifier rather than a forcing agent, since its atmospheric concentration is controlled by temperature rather than direct human emissions. Teacher note: Correct the common student misconception that CO_2 is the 'most powerful' greenhouse gas — it is the most significant forcing agent due to its long residence time and rising concentration, but CH_4 and N_2O have higher per-molecule warming potential.	not only by CO_2 from industrial nations but also by local CH_4 and N_2O sources, meaning Kenya's own land-use choices have measurable climate implications. Teacher note: Frame this carefully to avoid burdening students with guilt; the goal is scientific agency, not blame.
Lesson 5 **How Do Human Activities Amplify the Greenhouse Effect — and What Does This Mean for Mount Kenya and Kenyan Farmers?**	Students examined historical CO_2 concentration data (ice-core records extending 800,000 years back, showing natural cycles between ~180–280 ppm) contrasted with the post-industrial rise to over 420 ppm today. They observed Kenyan-specific deforestation data from the Mau Forest complex and the contribution of charcoal production (used for cooking ugali and nyama choma in Nairobi) to national carbon emissions. A case study of a maize farmer in Nakuru described how shifted long-rain seasons (March–May) have caused crop failures. Teacher note: The ice-core graph is one of the most powerful data visualisations in climate science — give students time to sit with it and annotate it before offering interpretation. Ask: 'What do you notice? What do you wonder?'	Human activities — including burning fossil fuels (petrol, diesel, charcoal), deforestation of forests like the Mau and Karura, industrial processes, and agricultural practices — release additional CO_2, CH_4, and N_2O into the atmosphere at rates far exceeding natural carbon cycle fluxes. This enhanced greenhouse forcing raises global mean temperatures, intensifies the hydrological cycle unevenly, and disrupts the atmospheric circulation patterns (including the Inter-Tropical Convergence Zone, ITCZ) that govern Kenya's two rain seasons (long rains: March–May; short rains: October–December). Students learned to distinguish natural climate variability from anthropogenic forcing using signal-to-noise reasoning. Teacher note: The ITCZ connection to Kenyan rainfall is a critical local link — use a diagram showing how a warmer Indian Ocean surface shifts ITCZ positioning relative to Kenya.	Students can now provide a full, evidence-based causal explanation for both parts of the driving question: (1) Mount Kenya's glaciers are melting because anthropogenic greenhouse gas emissions have raised tropospheric temperatures above the energy balance that sustained the glaciers through the Holocene. (2) Kenyan farmers' rains are becoming unpredictable because enhanced greenhouse forcing alters sea-surface temperatures in the Indian Ocean, shifts the ITCZ, and intensifies or suppresses convective rainfall systems unevenly across Kenya's diverse agroecological zones. Teacher note: By Lesson 5 students should be able to narrate a complete causal chain from a Nairobi charcoal jiko to a failed maize harvest in Nakuru — use this as an informal formative assessment prompt.
Lesson 6 **The Ozone Layer: A Separate Shield — and How Its Damage Connects to Mount Kenya's Disappearing Ice**	Students examined satellite images of the Antarctic ozone hole and UV index data for Nairobi and Kisumu compared to pre-ozone-depletion baselines. They observed that ozone depletion (caused by chlorofluorocarbons, CFCs) is a chemically distinct process from the greenhouse effect, yet both involve altered atmospheric radiative properties. Data showed that the Montreal Protocol (1987) has successfully begun reversing ozone depletion — a rare positive case study in international	The ozone layer (located in the stratosphere, 15–35 km altitude) absorbs incoming UV-B radiation through a photochemical cycle involving O_2 and O_3 molecules; this absorption prevents UV-B from reaching Earth's surface where it would damage DNA and biological tissue. Ozone depletion is caused by catalytic destruction of O_3 by chlorine and bromine radicals derived from CFCs and HCFCs — molecules not primarily implicated in the greenhouse effect. However, a secondary	Students can now definitively separate the ozone layer problem from the greenhouse effect in their explanations — a key conceptual achievement of the unit. They can articulate that Mount Kenya's glacier retreat is driven by the enhanced greenhouse effect (IR trapping by CO_2, CH_4, etc.), not directly by the ozone hole (which affects UV-B, not heat retention). However, they can also explain the secondary connection: some ozone-depleting substitute chemicals are

	environmental policy. Teacher note: This lesson is specifically designed to correct the near-universal misconception that 'the hole in the ozone layer causes global warming.' Use a Venn diagram explicitly — the overlap is UV/IR radiation effects on surface temperature, but the mechanisms and gases involved are entirely different.	connection to Mount Kenya exists: some ozone-depleting substances (notably HCFCs and HFCs used as replacements) are potent greenhouse gases, creating a physical link between the two phenomena. Students also learned that stratospheric ozone depletion can alter stratospheric temperature profiles in ways that influence tropospheric circulation. Teacher note: The Montreal Protocol success story is pedagogically important — it demonstrates that physics-informed international policy can reverse atmospheric damage, providing a hopeful counterpoint to climate pessimism.	greenhouse gases, and stratospheric changes can feed back into surface climate. Teacher note: Update the DQB at the end of this lesson — cross off or annotate the question 'Is it the ozone hole that's melting the glacier?' which typically appears on students' initial DQBs, and celebrate the conceptual refinement this represents.
Lesson 7 **Responding to Climate Change — Mitigation, Adaptation, and the Physics of Solutions**	Students examined case studies of Kenyan climate responses: (1) the Kenya Meteorological Department's early warning systems for farmers in Siaya and Homa Bay counties; (2) agroforestry programmes in Kericho tea estates sequestering carbon; (3) Lake Turkana Wind Power — Africa's largest wind farm — displacing fossil fuel generation; (4) solar irrigation pumps replacing diesel pumps for vegetable farmers near Naivasha. Data on Kenya's nationally determined contributions (NDCs) under the Paris Agreement were reviewed, along with projected temperature scenarios for Mount Kenya under 1.5 °C vs. 2.0 °C warming pathways. Teacher note: This lesson should feel empowering, not overwhelming. Frame every case study around the physics learned in earlier lessons — students should be able to explain WHY each solution works using radiation budget, GWP, and carbon cycle concepts.	Mitigation strategies reduce the concentration of greenhouse gases in the atmosphere (e.g., renewable energy, reforestation, reduced deforestation) and are grounded in the physics of reducing radiative forcing. Adaptation strategies adjust human and ecological systems to tolerate the climate changes already locked in (e.g., drought-resistant maize varieties, shifted planting calendars, early warning systems). Students learned to evaluate solutions using physics criteria: a strategy's effectiveness scales with how much it reduces net greenhouse gas forcing (in W/m^2 or CO_2-equivalent tonnes). They also learned that even under optimistic mitigation scenarios, the Lewis Glacier on Mount Kenya is projected to disappear by 2040–2050 due to the thermal inertia of the climate system — committed warming already in the pipeline. Teacher note: The concept of committed warming (warming already guaranteed by past emissions, even if emissions stopped today) is emotionally challenging for students — handle with care and redirect toward the science of why thermal inertia exists (ocean heat uptake, long atmospheric residence times of CO_2).	Students can now construct a complete, physics-grounded explanation for the driving question: Mount Kenya's glaciers are melting and Kenyan farmers' rains are becoming unpredictable because anthropogenic greenhouse gas emissions have enhanced the natural radiative forcing of the atmosphere, raising surface temperatures and disrupting atmospheric circulation patterns. The physics of the atmosphere — radiation budgets, molecular IR absorption, the Stefan–Boltzmann relationship, GWP, and feedback mechanisms — explains both the problem and the levers available for response. Students' final models should show the full causal chain from emissions sources (charcoal jikos, deforestation, livestock) through atmospheric physics to observed impacts (glacier retreat, ITCZ shifts, crop failures) and back to mitigation and adaptation pathways. Teacher note: The final model revision is the summative performance task — assess whether each student's model correctly represents energy flow, identifies at least three greenhouse gases with their sources, distinguishes the greenhouse effect from ozone depletion, and proposes at least one mitigation and one adaptation strategy with a physics justification.